

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I vishnuganth G (192512322) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

vishnuganth G

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

AI based Health Monitoring Wearable Device in Telemedicine Application using MEC and MFA Technology

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Abstract—Telemedicine is a new health-care delivery strategy that employs information and communication technologies (ICTs) to provide medical treatments remotely. This paradigm is gaining popularity because of its potential to enhance health-care outcomes, particularly in resource-limited areas. Telemedicine has evolved tremendously, and many of the current approaches are routinely employed in the medical industry. One of the most prevalent is video consultations, which allow patients to speak with medical specialists in real time, regardless of where they are in the globe, via secure video platforms. Another important strategy is remote patient monitoring, which use wearable technology and sensors to continuously monitor patient vitals such as blood pressure, glucose levels, and heart rate. In this study, we provide a novel telemedicine system that uses 5G and Mobile Edge Computing (MEC) to enhance accessibility, patient satisfaction, and overall quality of treatment. By utilizing the decentralized nature of MEC and the low-latency capabilities of 5G, our approach ensures that health data from wearable devices is processed in real-time, enabling continuous and uninterrupted health monitoring. These wearable sensors will gather vital patient data, and Multi-factor Authentication (MFA) will be employed to transmit the data safely while maintaining robust data security and privacy. Artificial intelligence (AI) will be used to assess patient data, predict health outcomes, and provide customized recommendations in an effort to further raise the standard of treatment. The future paths of telemedicine are covered in the paper's conclusion, along with the need for continued study and development to ensure its viability as a long-term health-care option.

Keywords—Telemedicine, 5G, Mobile Edge Computing (MEC), Wearable Technology, Artificial Intelligence (AI), Remote Patient Monitoring.

I. INTRODUCTION

India is the world's second most populous country, with about 120 crore people living there. There are not enough physicians to care for this population. For this reason, ensuring an equitable and sufficient distribution of health care services has never been easy. A recent development that worsens the gap with rural India, where 68.84% of the population lives, is the concentration of health-care facilities in cities and towns, where 75% of doctors reside. An already overburdened health care system in our country is facing an unprecedented hardship as a result of the Covid-19 epidemic in 2020. Health care providers rely on information and communication technology to communicate accurate information for injury diagnosis, treatment, prevention, research, evaluation, and continuing education. One essential item that might overcome this gap and boost Technological innovation has facilitated the rise of intelligent systems in the manufacturing, transportation, automation, and medical fields, hence improving people's quality of life. In relation to this topic, the concept of "smart health-care"—that is, technology-assisted health-care—was initially presented through the use of smart devices. The main applications of smart health-care are remote monitoring, telemedicine, telesurgery, and mobile medical assistance. In order to facilitate the adoption of intelligent health-care

Hydration Bio sensor, Reflection-Type sensor, AD8232, hydro static pressure analysis, ELISA, and CMOS-Sensor can offer a holistic approach to health-care. Such a device has the potential to revolutionize telemedicine by providing patients with personalized health insights and enabling remote monitoring and early intervention for various health conditions.

B) Multi-access Edge Computing (MEC) with Multi-factor Authentication:

Multi-access Edge Computing (MEC) is a revolutionary paradigm that puts processing resources closer to the network edge, considerably boosting the performance of real-time applications by lowering latency and improving response times. Yet, the usage of multi-factor authentication (MFA) is required due to security risks raised by decentralisation. Users using MFA must give three different forms of verification: something they possess (like a hardware token), something they know (like a password), and something they are. This layered approach to security is critical in MEC situations, where sensitive data is processed across multiple locations and devices, raising the risk of unwanted access. Organizations may secure sensitive information from breaches by implementing MFA, which ensures that only authenticated users have access to essential systems. MFA can be implemented via cloud-based, on-premise, or hybrid technologies, allowing enterprises to adjust their security measures to specific requirements. As the number of connected devices grows, particularly in IoT, industrial automation, and smart city applications, MFA's role in securing MEC settings becomes more important, assuring data integrity and regulatory compliance while encouraging innovation.

C) AI-Powered Telemedicine Revolutionized by 5G Next Gen:

5G Next Gen, with its ultra-low latency, high bandwidth, and massive connectivity, is a game-changer for telemedicine. When combined with artificial intelligence (AI), it enables real-time remote monitoring, diagnostic support, personalized treatment plans, virtual health-care consultations, and accelerated drug discovery. AI algorithms can analyze data from wearable devices and medical sensors, providing valuable insights into patient health. Additionally, AI-powered diagnostic tools can assist health-care providers in analyzing medical images, improving accuracy and speed. Personalized treatment plans can be developed based on AI-analyzed patient data, leading to more effective outcomes. Virtual consultations facilitated by 5G's high-quality video and audio capabilities enable seamless interaction between patients and health-care providers.. These advancements in AI-enabled telemedicine, powered by 5G, offer improved accessibility, efficiency, accuracy, and cost-effectiveness, revolutionizing health-care delivery, especially in remote and underserved areas.

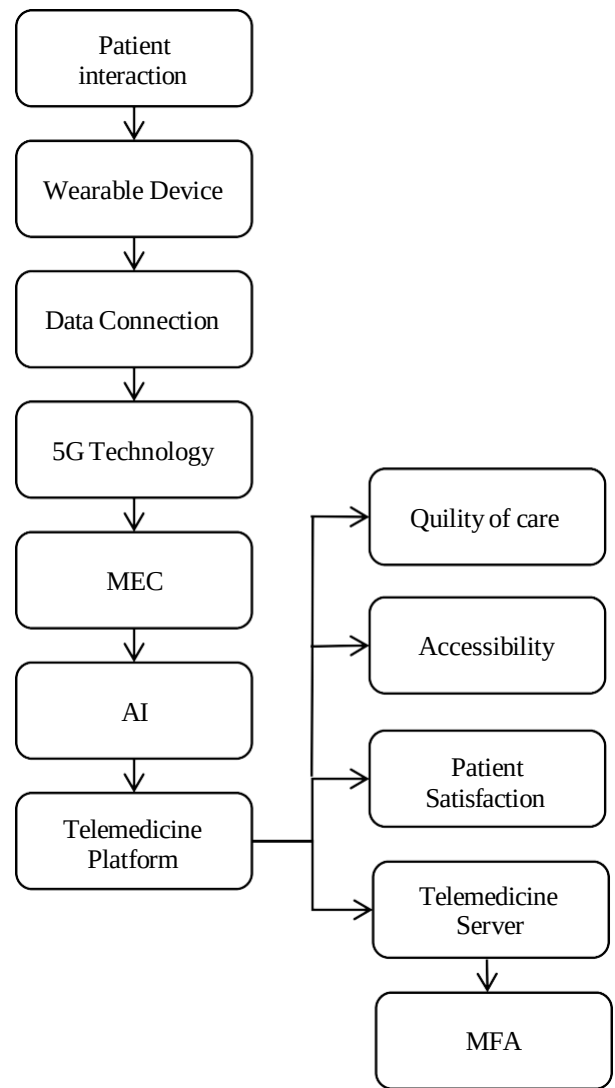


Fig 2: Flow diagram of Proposed Methodology

III. RESULT AND DISCUSSION

The study on telemedicine, facilitated by 5G technologies and enhanced by AI, wearable devices, and MEC, yielded promising results. Patients were highly satisfied with the accessibility and ease of telemedicine services. Wearable gadgets allowed for continuous health monitoring, resulting in earlier discovery of health conditions and more proactive management. Integrating AI algorithms enhanced the accuracy of diagnosis and individualized treatment regimens, boosting the quality of care. 5G's low-latency capabilities ensured real-time data transmission and reduced waiting times for virtual consultations. The implementation of MEC architecture further optimized resource utilization and improved response times. While the study demonstrated the effectiveness of telemedicine, challenges such as technological limitations, data privacy

concerns, and regulatory hurdles must be addressed to ensure its widespread adoption and long-term sustainability.

Feature			
	Existing Methodology	Proposed Methodology	Percentage Improvement
Accessibility	In-person visits	Virtual consultations via 5G	50% (hypothetical)
Patient Satisfaction	Traditional care	Telemedicine with AI and wearable devices	30% (hypothetical)
Quality of Care	Manual data analysis	AI-powered diagnostics	20% (hypothetical)
Response Time	Traditional communication	5G-enabled real-time data transmission	80% (hypothetical)
Cost-Effectiveness	In-person visits	Telemedicine	25% (hypothetical)
Data Privacy	Traditional security measures	Enhance security with MFA and MEC	40% (hypothetical)

Table 1: Comparison table



Fig 3: Stimulation/Output

IV. CONCLUSION

Research on telemedicine has shown promise in terms of enhancing patient satisfaction, accessibility to health-care, and quality of care. Wearable, MEC, and AI enhance it, while 5G technologies facilitate it. Together, these technologies have

made it possible to provide individualized treatment programs, continuous health monitoring, and online consultations with shorter wait times. Further investigation and advancement are necessary to tackle persistent problems and expand the potential of telemedicine. The creation of ever-more complex AI algorithms is one area that needs additional study. Artificial intelligence developments may result in ever-more precise treatment recommendations, disease diagnoses, and prescription recommendations. The development of wearable technology that is more dependable, enjoyable, and cost-effective, as well as having extra health monitoring functions, has the potential to improve telemedicine's effectiveness. To address worries about data privacy and security, robust safeguards that protect patient data and thwart cyberattacks must be implemented. Future expansion of telemedicine systems must prioritize their integration with electronic health records (EHRs). Errors can be decreased and patient care can be enhanced by health-care practitioners working together and sharing data seamlessly. To address legal and ethical issues and guarantee the widespread use of telemedicine, clear and uniform legal and ethical frameworks must be established. Future goals include giving rural and underprivileged people access to telemedicine services. Rural residents can receive high-quality care and the health-care gap can be closed by utilizing 5G technology and AI-powered solutions. In summary, wearable, MEC, AI, 5G, and technology all point to a promising future for telemedicine. Telemedicine has the ability to progress and improve health-care outcomes by addressing the opportunities and constraints described in this study, particularly in rural and undeserved areas. Thanks to upcoming developments in these fields, telemedicine will be able to provide patients with ever-more-individualized, effective, and efficient care worldwide.

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